

Four taxa and two areas

Now, we load the data included in the package: tree and distribution.

```
## Libraries
```

```
library(blepd)
```

```
## Loading required package: ape
```

```
## Loading required package: picante
```

```
## Loading required package: vegan
```

```
## Loading required package: permute
```

```
## Loading required package: lattice
```

```
## This is vegan 2.6-4
```

```
## Loading required package: nlme
```

```
library(ggtree)
```

```
## ggtree v3.10.1 For help: https://yulab-smu.top/treedata-book/
```

```
##
```

```
## If you use the ggtree package suite in published research, please cite  
## the appropriate paper(s):
```

```
##
```

```
## Guangchuang Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
```

```
## ggtree: an R package for visualization and annotation of phylogenetic
```

```
## trees with their covariates and other associated data. Methods in
```

```
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
```

```
##
```

```
## Shuangbin Xu, Lin Li, Xiao Luo, Meijun Chen, Wenli Tang, Li Zhan, Zehan
```

```
## Dai, Tommy T. Lam, Yi Guan, Guangchuang Yu. Ggtree: A serialized data
```

```
## object for visualization of a phylogenetic tree and annotation data.
```

```
## iMeta 2022, 1(4):e56. doi:10.1002/imt2.56
```

```
##
```

```
## Guangchuang Yu. Data Integration, Manipulation and Visualization of
```

```
## Phylogenetic Trees (1st edition). Chapman and Hall/CRC. 2022,
```

```
## doi:10.1201/9781003279242
```

```
##
```

```
## Attaching package: 'ggtree'
```

```
## The following object is masked from 'package:nlme':
```

```
##
```

```
## collapse
```

```
## The following object is masked from 'package:ape':
```

```
##
```

```
## rotate
```

```
library(ggplot2)
```

```
## Data
```

```
data(package = "blepd")
```

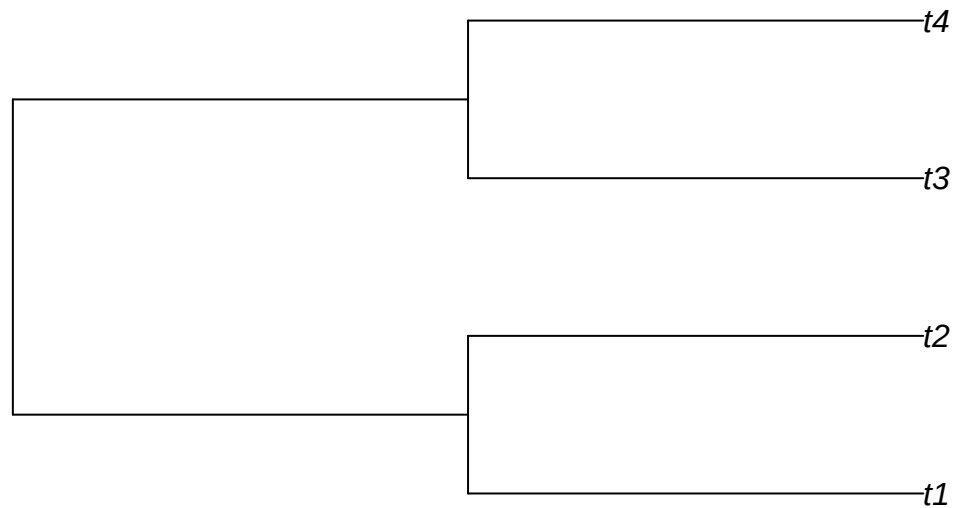
```
## trees
```

```
data(tree)
```

```
str(tree)
```

```
## List of 5
## $ edge      : int [1:6, 1:2] 5 6 6 5 7 7 6 1 2 7 ...
## $ edge.length: num [1:6] 1 1 1 1 1 1
## $ Nnode      : int 3
## $ tip.label  : chr [1:4] "t1" "t2" "t3" "t4"
## $ root.edge  : num 1
## - attr(*, "class")= chr "phylo"
## - attr(*, "order")= chr "cladewise"
```

```
plot(tree,use.edge.length =T)
```



```
initialTree <- tree
```

```
## distributions
```

```
data(distribution)
```

```
str(distribution)
```

```
## int [1:2, 1:4] 1 0 0 1 1 0 0 1
## - attr(*, "dimnames")=List of 2
## ..$ : chr [1:2] "A1" "A2"
```

```

##   ..$ : chr [1:4] "t1" "t2" "t3" "t4"
dist4taxa <- distribution

## distribution to XY to be able to plot the data, otherwise skip the next step

distXY <- matrix2XY(dist4taxa)

distXY

##   Terminal Area
## 1      t1    A1
## 2      t3    A1
## 3      t2    A2
## 4      t4    A2

## plotting

## the tree

plotTree <- ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=6, color="black") +
  theme_tree2() +
  ggtitle("Four terminals, equal branch length")

##print(plotTree)

## the distribution

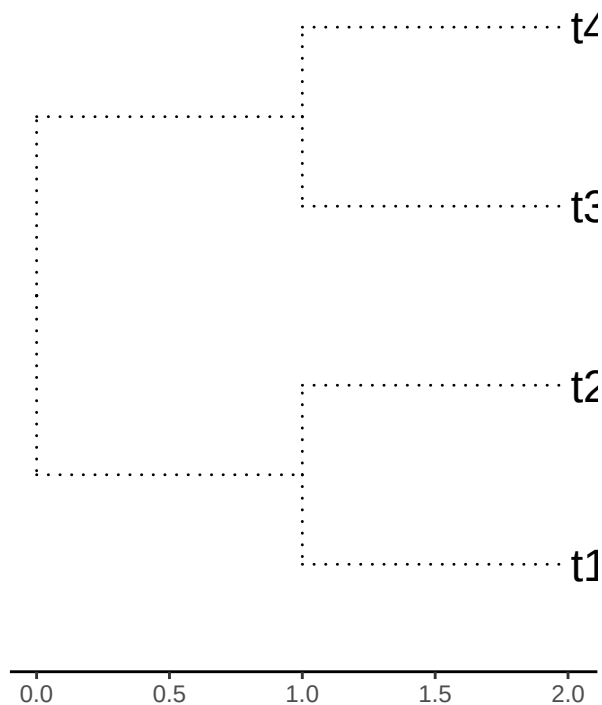
plotDistrib <- ggplot(data=distXY,
                      aes(x= Area, y= Terminal , size =150)) +
  geom_point() +
  theme_bw() +
  theme( axis.text.y = element_blank(),
        panel.grid.major = element_blank(), # Remove major gridlines
        panel.grid.minor = element_blank(), # Remove minor gridlines
        axis.line = element_line(colour = NA_character_), # Remove axis lines
        axis.ticks = element_blank(), # Remove axis ticks
        legend.position = "none"
        ) +
  labs(title = "Distributions",
       y = "",
       x = "Area")

##print(plotDistrib)

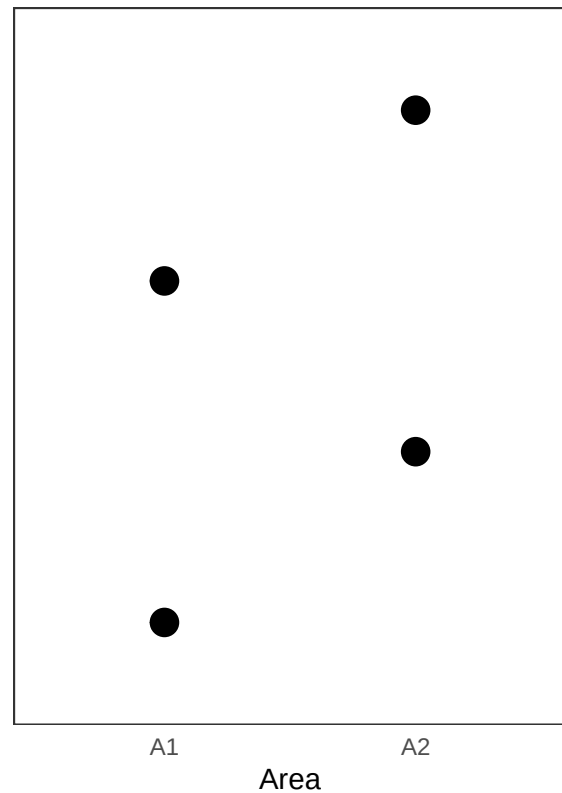
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)

```

Four terminals, equal branch length



Distributions



We check whether names in both objects: initialTree and dist4taxa, are the same.

```
all(colnames(dist4taxa) == initialTree$tip.label)
```

```
## [1] TRUE
```

We report the branch length, and calculate the PD values.

```
initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)
```

```
initialPD
```

```
## [1] 4 4
```

As expected there is a tie between both areas.

Now, we can see the impact of the possible null models

```
for( modelo in c("simpleswap", "allswap", "uniform") ){
  for( rama in c("terminals", "internals", "all") ){
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
```

```

    )

    cat( "\n\t Model=",modelo,
         "\n\t Branchs swapped=",rama,"\n" )

    printBlepd(val)

    cat( "\n\n\n" )

}

}

```

```

## model to test simpleswap reps 100
##
##   Model= simpleswap
##   Branchs swapped= terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
##   Model= simpleswap
##   Branchs swapped= internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
##   Model= simpleswap
##   Branchs swapped= all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test allswap reps 100
##
##   Model= allswap
##   Branchs swapped= terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100

```

```

##
##
##
## model to test allswap reps 100
##
##   Model= allswap
##   Branchs swapped= internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test allswap reps 100
##
##   Model= allswap
##   Branchs swapped= all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test uniform reps 100
##
##   Model= uniform
##   Branchs swapped= terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test uniform reps 100
##
##   Model= uniform
##   Branchs swapped= internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test uniform reps 100
##
##   Model= uniform
##   Branchs swapped= all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent

```

```
## 1          A1A2  100    100
```

As expected, as all branches are EQUAL nothing happens

Now, we test if the number of replicates has any effect.

```
for(repetir in 1:2){  
  
    val <- swapBL( tree=initialTree,  
                  distribution = dist4taxa,  
                  model = "allswap",  
                  branch = "terminals",  
                  nTimes = 10**repetir  
                )  
  
    printBlepd(val)  
  
}
```

```
## model to test allswap reps 10  
##  
## BestInitial:A1A2  
##   AreaSelected Freq Percent  
## 1          A1A2   10     100  
## model to test allswap reps 100  
##  
## BestInitial:A1A2  
##   AreaSelected Freq Percent  
## 1          A1A2  100     100
```

Roughly speaking, from 1000 on the results are alike, and the largest difference is using 10 or 100 replicates. As a rule of thumb, we must use at least 100 replicates, but 1000 will be better.

Now, let us see if the difference in results could be assigned to the longest branches (or not).

```
## we could use a single command (*evalBranch*)  
  
AllTerminals <- evalBranch(tree=initialTree,  
                           distribution = dist4taxa,  
                           branchToEval="terminals",  
                           approach="all")
```

```
## Loading required package: maps  
##  
## Attaching package: 'phytools'  
## The following object is masked from 'package:vegan':  
##  
##     scores
```

```
print.evalBranchAll(AllTerminals)
```

```
##  
## BranchNumber initArea    Mod-upper   Delta%  Mod-lower   Delta%  
## NA   A1A2     A1   2.01     A2   -1  
## NA   A1A2     A2   2.01     A1   -1  
## NA   A1A2     A1   2.01     A2   -1  
## NA   A1A2     A2   2.01     A1   -1
```

```
##
```

```
AllTerminals <- evalBranch(tree=initialTree,  
                           distribution = dist4taxa,  
                           branchToEval="internals",  
                           approach="all")
```

```
print.evalBranchAll(AllTerminals)
```

```
##
```

```
## BranchNumber initArea    Mod-upper  Delta%  Mod-lower  Delta%  
## 6    A1A2    A1A2    0    A1A2    0  
## 7    A1A2    A1A2    0    A1A2    0  
##
```

```
AllInternals <- evalBranch(tree=initialTree,  
                           distribution = dist4taxa,  
                           branchToEval="all",  
                           approach="all")
```

```
print.evalBranchAll(AllInternals)
```

```
##
```

```
## BranchNumber initArea    Mod-upper  Delta%  Mod-lower  Delta%  
## 6    A1A2    A1A2    0    A1A2    0  
## NA   A1A2    A1    2.01    A2    -1  
## NA   A1A2    A2    2.01    A1    -1  
## 7    A1A2    A1A2    0    A1A2    0  
## NA   A1A2    A1    2.01    A2    -1  
## NA   A1A2    A2    2.01    A1    -1  
##
```

This is the same